

Missing Numbers - CSES

Introductory Problems

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1. Problem

You are given all numbers between $1, 2, \dots, n$ except one. Your task is to find the missing number.

- **Input:**

The first input line contains an integer n . The second line contains $n - 1$ numbers. Each number is distinct and between 1 and n (inclusive).

- **Output:**

Print the missing number.

- **Constraints:** $2 \leq n \leq 2 \cdot 10^5$

2. Solution

As every value but one is given, the sum of all of them, given the j -number is missing will be as follows:

$$\sum_{i \leq n, i \neq j} x_i = \sum_{i \leq n, i \neq j} i$$

Also the gaussian sum already gives us a way to compute easily the sum of a range of numbers, in this case:

$$\sum_{i \leq n} x_i = \frac{n(n-1)}{2}$$

Thus we only need to store the sum of the given numbers and subtract it from the gaussian sum of the values from 1 to n .

$$x_j = \frac{n(n-1)}{2} - \sum_{i \leq n, i \neq j} x_i$$

As the constraint of n is $2 \cdot 10^5$, the gaussian sum will exceed clearly what can be stored in a 32b integer ($n^2 \approx 4 \cdot 10^{10}$), so a 64b integer (`long long int`) is mandatory.

Now, the computation of the missing number is in constant time, the only obstacle is the reading of the input of n values in total, so the program has a complexity $O(n)$.

3. Code

```
#include<bits/stdc++.h>
using namespace std;
using ll = long long int;

int main(){
    ll n, m, aux = 0, total;
    cin >> n;
    total = n * (n + 1) / 2;
    for (int i = 0; i < n-1; i++){
        cin >> m;
        aux += m;
    }
    printf("%lld\n", total - aux);
    return 0;
}
```